



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2007

HIGHER SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 4: TRIAL HSC

Mathematics Extension 2

TIME ALLOWED: 3 HOURS

(PLUS 5 MINUTES READING TIME)

Outcomes Assessed	Questions	Marks
Determines the important features of graphs of a wide variety of functions, including conic sections	3, 5	
Applies appropriate algebraic techniques to complex numbers and polynomials	2, 4	
Applies further techniques of integration, such as slicing and cylindrical shells, integration by parts and recurrence formulae, to problems	1, 6	
Synthesises mathematical solutions to harder problems and communicates them in an appropriate form	7, 8	

Question	1	2	3	4	5	6	7	8	Total	%
Marks	/15	/15	/15	/15	/15	/15	/15	/15	/120	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started on a new page

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Total marks – 120
Attempt Questions 1 – 8
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1. (15 marks) Use a SEPARATE writing booklet.	Marks
(a) Find $\int \cos^5 x \sin x \, dx$.	2
(b) Find $\int \frac{2x}{\sqrt{x^2 - 4}} \, dx$ using the substitution $u = x^2 - 4$.	3
(c) (i) Express $\frac{2 - x^2}{(x^2 + 1)(x^2 + 4)}$ as a sum of partial fractions.	2
(ii) Hence show that $\int_0^3 \frac{2 - x^2}{(x^2 + 1)(x^2 + 4)} \, dx = \tan^{-1}\left(\frac{3}{11}\right)$.	4
(d) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 - \sin x}$.	4

Question 2. (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Let $z = 1 + i$ and $w = 1 - 2i$. Find in the form $x + iy$,

(i) $z\bar{w}$ **1**

(ii) $3z + iw$ **1**

(iii) $\frac{w}{z}$ **1**

(b) Let $\beta = -1 + i$

(i) Express β in modulus-argument form. **2**

(ii) Express β^4 in modulus-argument form. **1**

(iii) Hence evaluate β^{20} **1**

(c) (i) Sketch, on the same Argand diagram, the locus specified by, **4**

1. $|z - 9| = |z + 1|$

2. $|z - 2 + i| = 2$

(ii) Hence write down all the values of z which satisfy simultaneously **1**

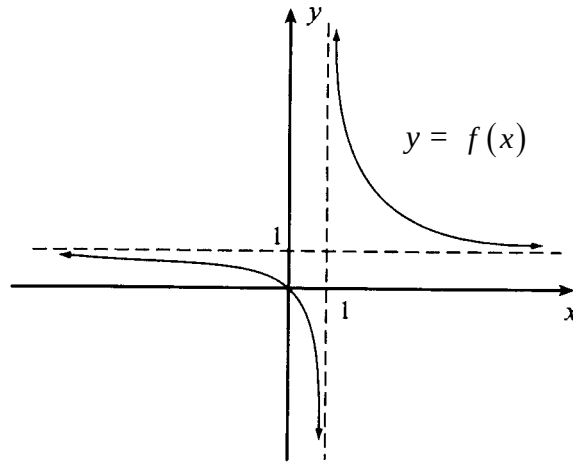
$$|z - 9| = |z + 1| \quad \text{and} \quad |z - 2 + i| = 2$$

(d) Prove $|z_1 - z_2|^2 + |z_1 + z_2|^2 = 2|z_1|^2 + 2|z_2|^2$ and interpret this result geometrically. **3**

Question 3. (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The diagram bellows shows the graph of $y = f(x)$

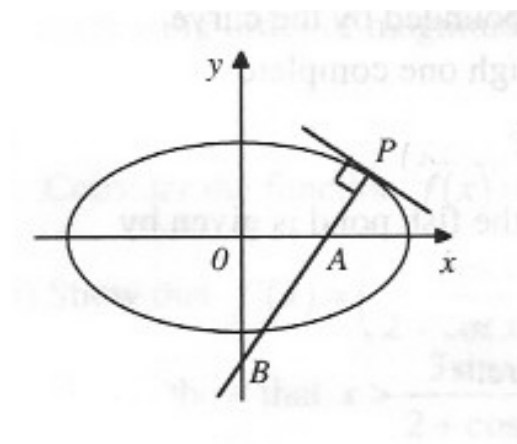


Draw separate one-third page sketches of the graphs of the following:

- | | | |
|-------|--------------------|---|
| (i) | $y = f(x-1) - 1$ | 2 |
| (ii) | $y = f(x) $ | 2 |
| (iii) | $y = e^{f(x)}$ | 2 |
| (iv) | $y = \log_e(f(x))$ | 2 |

(Question 3 continues over)

- (b) $P(a \cos \theta, b \sin \theta)$, where $0 < \theta < \frac{\pi}{2}$, is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b > 0$.



The normal at P cuts the x axis at A and the y axis at B .

- (i) Show that the normal at P has the equation 2

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

- (ii) Show that triangle OAB has areas $\frac{(a^2 - b^2)^2 \sin \theta \cos \theta}{2ab}$ 2

- (iii) Find the maximum area of the triangle OAB and the coordinates of P when this maximum occurs. 3

- (a) Given that α , β and γ are the roots to the equation $x^3 - x^2 + 5x - 3 = 0$, find the equation whose roots are $\alpha\beta$, $\alpha\gamma$ and $\beta\gamma$ **3**

- (b) Let α be the complex root of the polynomial $z^7 = 1$ with the smallest possible argument.

Let $\theta = \alpha + \alpha^2 + \alpha^4$ and $\phi = \alpha^3 + \alpha^5 + \alpha^6$

- (i) Explain why $\alpha^7 = 1$ and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$ **2**

- (ii) Show $\theta + \phi = -1$ and $\theta\phi = 2$ **3**

Hence write a quadratic equation whose roots are θ and ϕ

- (iii) Show that $\theta = -\frac{1}{2} + \frac{i\sqrt{7}}{2}$ and $\phi = -\frac{1}{2} - \frac{i\sqrt{7}}{2}$ **2**

- (iv) Write α in modulus argument form and show **2**

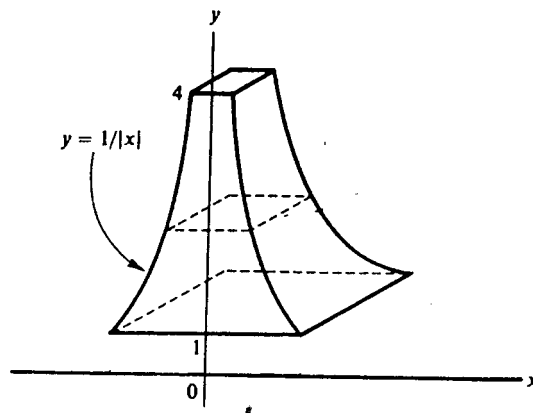
$$\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2} \quad \text{and} \quad \sin \frac{4\pi}{7} + \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$$

- (c) The polynomial $P(z)$ is defined by $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$. **3**

Given that $z - 2 + i$ is a factor of $P(z)$, express $P(z)$ as a product of real quadratic factors.

- (a) Consider the curve given by $5y - xy = x^2 - x - 2$
- (i) Show that the curve has stationary points at $5 \pm 3\sqrt{2}$ 2
- (ii) Explain why the curve approaches that of $y = -x - 4$ as $x \rightarrow \pm \infty$ 2
- (b) For the hyperbola $\frac{x^2}{4} - \frac{y^2}{5} = 1$, find
- (i) The eccentricity. 1
- (ii) The coordinates of the foci. 1
- (iii) The equations of the directrices. 1
- (iv) The equations of the asymptotes. 1
- (v) Sketch the hyperbola indicating the foci, the directrices and the asymptotes. 1
- (vi) Show that the point $P(2 \sec \theta, \sqrt{5} \tan \theta)$ lies on the hyperbola and prove that the tangent to the hyperbola at P has the equation 2
- $$\frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{5}} = 1$$
- (vii) If the tangent at P cuts the asymptotes at L and M , prove that $LP = PM$ and the area of triangle OLM is independent of the position of P . 4

- (a) The plan of a steeple is bounded by the curve $y = \frac{1}{|x|}$ and the lines $y = 4$ and $y = 1$.

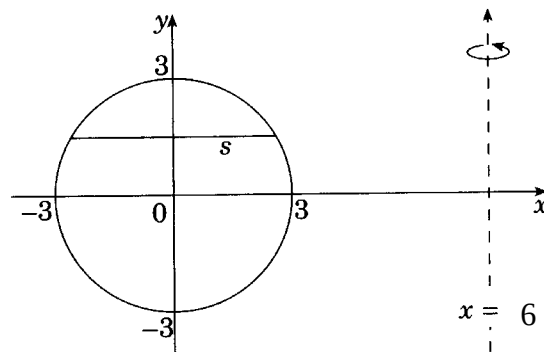


Each horizontal cross-section is a square.

Find the volume of the steeple.

4

- (b) The circle $x^2 + y^2 = 9$ is rotated about the line $x = 6$ to form a ring.



- (i) When the circle is rotated, the line segment S at height y sweeps out an annulus.

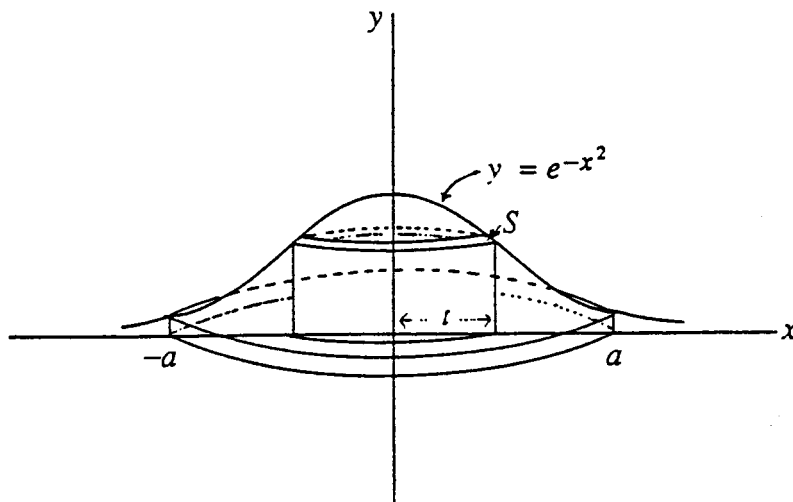
2

Find the area of the Annulus.

- (ii) Hence find the volume of the ring

3

- (c) The region under the curve $y = e^{-x^2}$ and above the x-axis is rotated about the y axis for $-a \leq x \leq a$ to form a solid as shown below.



- (i) Divide the resulting solid into cylindrical shells S of radius t as shown in the diagram and show each shell S has an approximate volume given by $\delta V = 2\pi t e^{-t^2} \delta t$, where δt is the thickness of the shell. 2
- (ii) Hence calculate the volume of the solid. 2
- (iii) What is the limiting value of the volume of the solid as $a \rightarrow \infty$? 2

Question 7.

(15 marks) Use a SEPARATE writing booklet.

(a) Let $I_n = \int_0^1 (1-x^2)^n dx$.

(i) Show by using integration by parts $I_n = \frac{2n}{2n+1} I_{n-1}$ for $n = 0, 1, 2, 3, \dots$ **3**

(ii) Hence evaluate $\int_0^1 (1-x^2)^4 dx$ **3**

(b) A special dish is designed by rotating the region bounded by the curve $y = 2 \cos x$ ($0 \leq x \leq 2\pi$) and the line $y = 2$ through 360° about the y axis.

i) Use the method of cylindrical shells to show that the volume of the dish is given by **3**

$$4\pi \int_0^{2\pi} x(1 - \cos x) dx.$$

ii) Hence find the volume. **3**

(c) The polynomial $P(x)$ is given by $P(x) = 2x^3 - 9x^2 + 12x - k$, where k is real. **3**

Find the range of values for k for which $P(x) = 0$ has 3 real roots.

Question 8. (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Use integration by parts to find $\int \sin^{-1} x \, dx$. 3

(b) (i) Use De Moivre's Theorem to show that $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$ 3

(iv) Show that the equation $16x^4 - 16x^2 + 1 = 0$ has roots 3

$$x_1 = \cos \frac{\pi}{12}, x_2 = -\cos \frac{\pi}{12}, x_3 = \cos \frac{5\pi}{12}, x_4 = -\cos \frac{5\pi}{12}$$

(iii) Hence show that $\cos \frac{\pi}{12} = \frac{\sqrt{2+\sqrt{3}}}{2}$ 2

(c) $P(x)$ is a polynomial of degree n with rational coefficients. 4

If the leading coefficient is a_0 and $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are the roots of $P(x) = 0$ prove that:

$$P'(x) = \frac{P(x)}{x - \alpha_1} + \frac{P(x)}{x - \alpha_2} + \frac{P(x)}{x - \alpha_3} + \dots + \frac{P(x)}{x - \alpha_n}$$



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Mathematics Extension 2

Solutions

Question 1. (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int \cos^5 x \sin x \, dx$.

Solution

$$= -\frac{1}{6} \cos^6 x + C$$

2	Correct Solution
1	Omitting constant or negative sign

(b) Find $\int \frac{2x}{\sqrt{x^2-4}} \, dx$ using the substitution $u = x^2 - 4$.

Solution

$$\begin{aligned} u &= x^2 - 4 \\ \frac{du}{dx} &= 2x \\ dx &= \frac{du}{2x} \end{aligned} \qquad \begin{aligned} \int \frac{2x}{\sqrt{x^2-4}} \, dx &= \int \frac{2x}{\sqrt{u}} \frac{du}{2x} \\ &= \int \frac{1}{\sqrt{u}} \, du \\ &= 2\sqrt{u} + C \\ &= 2\sqrt{x^2-4} + C \end{aligned}$$

3	Correct Solution
2	Showing $\int \frac{1}{\sqrt{u}} \, du = 2\sqrt{u} + C$
1	Finding $dx = \frac{du}{2x}$

(c) (i) Express $\frac{2-x^2}{(x^2+1)(x^2+4)}$ as a sum of partial fractions.

Solution

$$\frac{2-x^2}{(x^2+1)(x^2+4)} = \frac{ax+b}{x^2+1} + \frac{cx+d}{x^2+4}$$

2	Correct Solution
1	Finding the expansion or arithmetic error

$$2-x^2 = (x^2+4)(ax+b) + (x^2+1)(cx+d)$$

$$2-x^2 = ax^3 + bx^2 + 4ax + 4b + cx^3 + dx^2 + cx + d$$

Equating co-efficients & solving simultaneously $a=0, b=1, c=0, d=-2$

$$\frac{2-x^2}{(x^2+1)(x^2+4)} = \frac{1}{x^2+1} - \frac{2}{x^2+4}$$

(ii) Hence show that $\int_0^3 \frac{2-x^2}{(x^2+1)(x^2+4)} dx = \tan^{-1}\left(\frac{3}{11}\right).$

Solution

$$\begin{aligned} \int_0^3 \frac{2-x^2}{(x^2+1)(x^2+4)} dx &= \int_0^3 \frac{1}{x^2+1} dx - \int_0^3 \frac{2}{x^2+4} dx \\ &= \left[\tan^{-1} x \right]_0^3 - 2 \int_0^3 \frac{1}{4+x^2} dx \\ &= \tan^{-1} 3 - 2 \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^3 \\ &= \tan^{-1} 3 - \tan^{-1} \frac{3}{2} \end{aligned}$$

Let $x = \tan^{-1} 3$ $y = \tan^{-1} \frac{3}{2}$

$\tan x = 3, \quad \frac{3}{2} = \tan y$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$= \frac{3 - \frac{3}{2}}{1 + 3 \cdot \frac{3}{2}}$$

$$\tan(x-y) = \frac{3}{11}$$

$$x-y = \tan^{-1} \frac{3}{11}$$

$$\tan^{-1} 3 - \tan^{-1} \frac{3}{2} = \tan^{-1} \frac{3}{11}$$

4	Correct Solution
3	One arithmetic error with correct process
2	Finding $\int_0^3 \frac{2-x^2}{(x^2+1)(x^2+4)} dx = \tan^{-1} 3 - \tan^{-1} \frac{3}{2}$
1	Finding $\left[\tan^{-1} x \right]_0^3 - 2 \int_0^3 \frac{1}{4+x^2} dx$

(d) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 - \sin x}$.

Solution

$$\begin{aligned}
 t &= \tan \frac{x}{2} & \int_0^{\frac{\pi}{2}} \frac{dx}{2 - \sin x} &= \int_0^1 \frac{\frac{2dt}{1+t^2}}{2 - \frac{2t}{1+t^2}} \\
 \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} & &= \int_0^1 \frac{\frac{2dt}{1+t^2}}{\frac{2+2t^2-2t}{1+t^2}} \\
 \frac{dt}{dx} &= \frac{1}{2} (1+t^2) & &= \int_0^1 \frac{1}{t^2 - t + 1} dt \\
 dx &= \frac{2dt}{1+t^2} & &= \int_0^1 \frac{1}{\left(t^2 - t + \frac{1}{4}\right) + \frac{3}{4}} dt \\
 x = \frac{\pi}{2}, t &= \tan \frac{\pi/2}{2} = 1 & &= \int_0^1 \frac{1}{\left(t^2 - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dt \\
 x = 0, t &= \tan \frac{\pi/2}{2} = 0 & &= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{2\left(t - \frac{1}{2}\right)}{\sqrt{3}} \right]_0^1 \\
 & & &= \frac{2\pi}{\sqrt{3}} \left(\frac{1}{3} \right)
 \end{aligned}$$

- | | |
|---|---|
| 4 | Correct Solution |
| 3 | One error with correct process |
| 2 | Finding $\int_0^1 \frac{1}{t^2 - t + 1} dt$ |
| 1 | Finding $dx = \frac{2dt}{1+t^2}$ |

(a) Let $z = 1 + i$ and $w = 1 - 2i$. Find in the form $x + iy$,

(i) $z\bar{w}$

Solution

$$\begin{aligned} z\bar{w} &= (1+i)(1+2i) \\ &= -1 + 3i \end{aligned}$$

1 Correct Solution

(ii) $3z + iw$

Solution

$$\begin{aligned} 3z + iw &= 3 + 3i + i(1 - 2i) \\ &= 5 + 4i \end{aligned}$$

1 Correct Solution

(iii) $\frac{w}{z}$

Solution

$$\begin{aligned} \frac{w}{z} &= \frac{1-2i}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{-1}{2} - \frac{3}{2}i \end{aligned}$$

1 Correct Solution

(b) Let $\beta = -1 + i$

(i) Express β in modulus-argument form.

Solution

$$|\beta| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2} \quad \arg \beta = \tan^{-1}\left(\frac{-1}{1}\right), \text{ in second quad. } = \frac{3\pi}{4}$$

$$\beta = \sqrt{2} \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$$

2 Correct Solution
1 Finding either $|\beta|$ or $\arg \beta$

(ii) Express β^4 in modulus-argument form.

Solution

$$\beta^4 = \left[\sqrt{2} \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right) \right]^4$$

1	Correct Solution
---	------------------

$$= 4(\cos(3\pi) + i \sin(3\pi)) \quad \text{by De Moivre's theorem}$$

(iii) Hence evaluate β^{20}

Solution

$$\beta^{20} = (\beta^4)^5$$

$$= \left(4(\cos(3\pi) + i \sin(3\pi)) \right)^5$$

$$= (-4)^5$$

$$= -1024$$

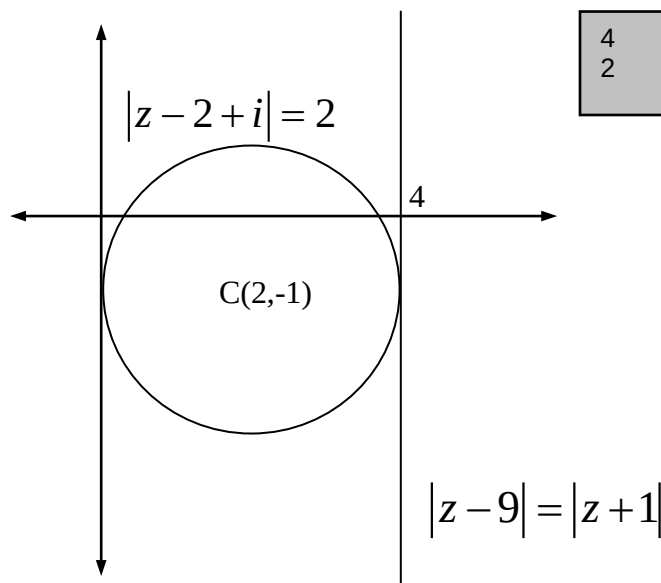
1	Correct Solution
0	If student did not interpret HENCE and reapplied De Moivre's Theorem

(c) (i) Sketch, on the same Argand diagram, the locus specified by,

1. $|z - 9| = |z + 1|$

2. $|z - 2 + i| = 2$

Solution



4	Correct Solution
2	For one correct solution

(ii) Hence write down all the values of z which satisfy simultaneously

$$|z - 9| = |z + 1| \quad \text{and} \quad |z - 2 + i| = 2$$

Solution

$$z = 4 - i$$

1	Correct Solution
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(d) Prove $|z_1 - z_2|^2 + |z_1 + z_2|^2 = 2|z_1|^2 + 2|z_2|^2$ and interpret this result geometrically.

Solution

Using the property $|w|^2 = w\bar{w}$

$$\begin{aligned}
 \text{LHS} &= |z_1 - z_2|^2 + |z_1 + z_2|^2 \\
 &= (z_1 - z_2)(\overline{z_1 - z_2}) + (z_1 + z_2)(\overline{z_1 + z_2}) \\
 &= (z_1 - z_2)(\overline{z_1} - \overline{z_2}) + (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\
 &= z_1\overline{z_1} - z_1\overline{z_2} - z_2\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2} \\
 &= 2z_1\overline{z_1} + 2z_2\overline{z_2} \\
 &= 2|z_1|^2 + 2|z_2|^2 \\
 &= \text{RHS}
 \end{aligned}$$

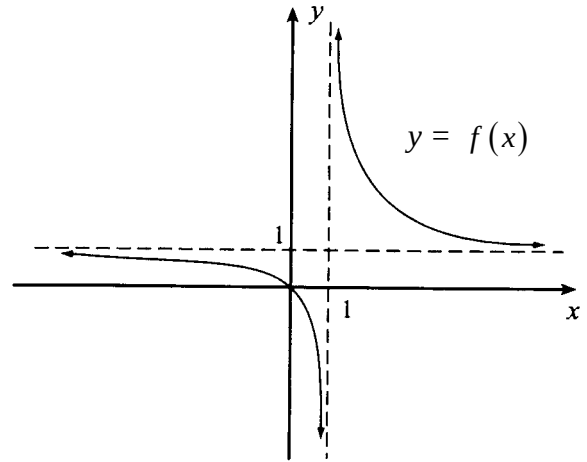
Geometric interpretation

Since $z_1 - z_2$ and $z_1 + z_2$ are diagonals of a parallelogram formed by opposite vertices z_1 and z_2 we can say

The sum of the diagonals squared is equal to 2 times the sum of adjacent sides squared.

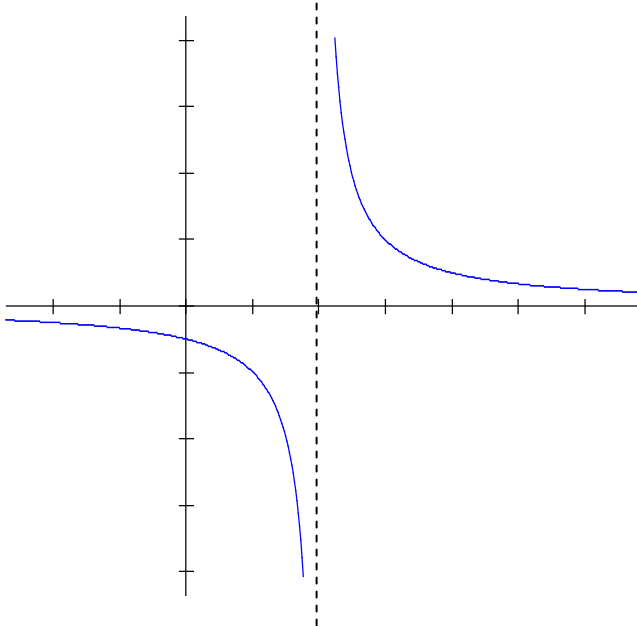
3	Correct Solution
2	Showed identity without geometric interpretation.
1	Demonstrating $ w ^2 = w\bar{w}$ or $\overline{z_1 \pm z_2} = \overline{z_1} \pm \overline{z_2}$

- (a) The diagram bellows shows the graph of $y = f(x)$



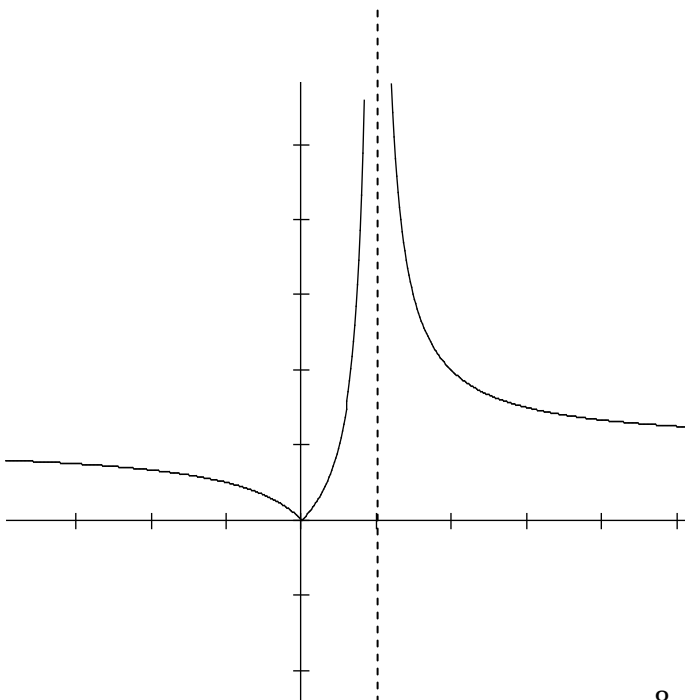
Draw separate one-third page sketches of the graphs

(i) $y = f(x-1) - 1$



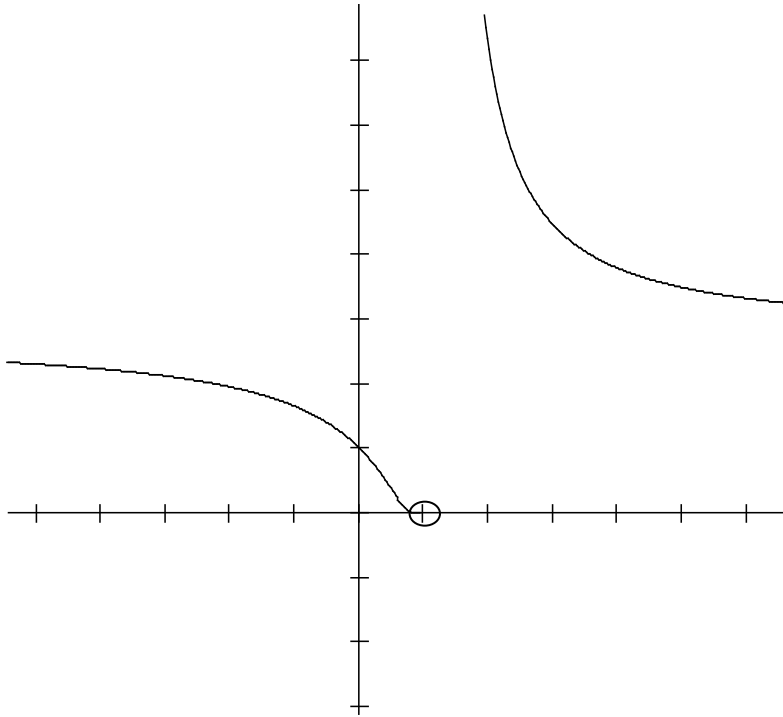
2	Correct Solution
1	For one correct translation only

ii) $y = |f(x)|$



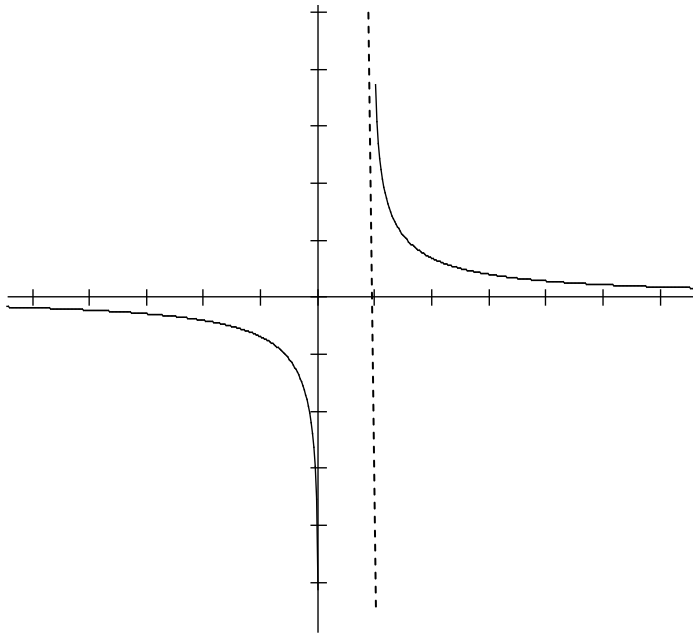
2	Correct Solution
1	Answer sans horizontal asymptote

(iii) $y = e^{f(x)}$



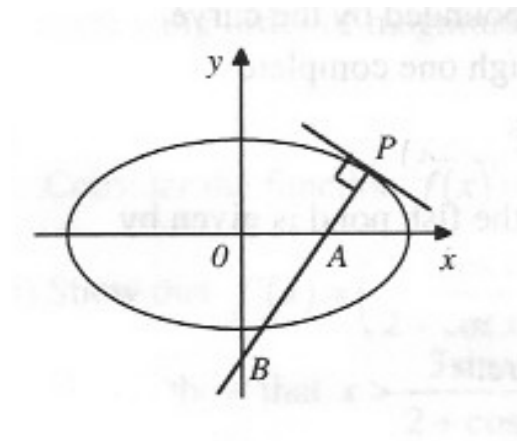
2	Correct Solution
1	For partially correct shape

(iv) $y = \log_e(f(x))$



2	Correct Solution
1	For partially correct shape

- (b) $P(a \cos \theta, b \sin \theta)$, where $0 < \theta < \frac{\pi}{2}$, is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b > 0$.



The normal at P cuts the x axis at A and the y axis at B .

- (i) Show that the normal at P has the equation

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

Solution

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2} \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{2x^2}{a^2} \times \frac{b^2}{2y^2}$$

$$\frac{dy}{dx} = -\frac{b^2}{a^2} \times \frac{x}{y}$$

so the gradient of the tangent at $P(x_1, y_1)$ is $-\frac{b^2}{a^2} \times \frac{x_1}{y_1}$

The gradient of the normal is $-\frac{b^2}{a^2} \times \frac{x_1}{y_1} \times m = -1$

$$m = \frac{a^2}{b^2} \times \frac{y_1}{x_1}$$

$$m = \frac{a \sin \theta}{b \cos \theta}$$

So the equation of the normal is

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$by - b^2 \sin \theta = a \frac{\sin \theta}{\cos \theta} (x - a \cos \theta)$$

$$\frac{by}{\sin \theta} - b^2 = \frac{ax}{\cos \theta} - a^2$$

2	Correct Solution
1	Finding gradient of the tangent

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

(ii) Show that triangle OAB has areas $\frac{(a^2 - b^2)^2 \sin \theta \cos \theta}{2ab}$

Solution

For point A, $y = 0$

$$ax \sin \theta - b(0) \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

$$x = \frac{(a^2 - b^2) \cos \theta}{a}$$

For point B, $x = 0$

$$a(0)\sin\theta - by\cos\theta = (a^2 - b^2)\sin\theta\cos\theta$$

$$y = \frac{(a^2 - b^2)\sin\theta}{-b}$$

$$\begin{aligned} \text{Area AOB} &= \frac{1}{2} \times \left| \frac{(a^2 - b^2)\cos\theta}{a} \right| \times \left| \frac{(a^2 - b^2)\sin\theta}{-b} \right| \\ &= \frac{(a^2 - b^2)^2 \sin\theta\cos\theta}{2ab} \quad \text{since } a > b \\ &= \frac{(a^2 - b^2)^2 \sin\theta\cos\theta}{2ab} \times \frac{2}{2} \\ &= \frac{(a^2 - b^2)^2 2\sin\theta\cos\theta}{4ab} \\ &= \frac{(a^2 - b^2)^2 \sin 2\theta}{4ab} \end{aligned}$$

2	Correct Solution
1	Finding x and y coordinates

- (iii) Find the maximum area of the triangle OAB and the coordinates of P when this maximum occurs.

Solution

$$\begin{aligned} \frac{dA}{d\theta} &= \frac{(a^2 - b^2)^2}{4ab} \cos 2\theta \times 2 \\ &= \frac{(a^2 - b^2)^2 \cos 2\theta}{2ab} \end{aligned}$$

Finding turning pts

$$\frac{(a^2 - b^2)^2 \cos 2\theta}{2ab} = 0$$

Alternatively

$$\frac{(a^2 - b^2)^2 \sin 2\theta}{4ab} \text{ is a maximum when } \sin 2\theta = 1$$

$$\theta = \frac{\pi}{4}, \quad \text{since } 0 < \theta < \frac{\pi}{2}$$

$$(a^2 - b^2)^2 \cos 2\theta = 0$$

$$\cos 2\theta = 0$$

$$2\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{4}$$

Testing for max.

$$\frac{d^2 A}{d\theta^2} = \frac{(a^2 - b^2)^2}{2ab} (-\sin 2\theta) \times 2$$

$$= -\frac{(a^2 - b^2)^2 \sin 2\theta}{ab}$$

$$< 0 \quad \text{since } a > b \text{ and } 0 < \theta < \frac{\pi}{2}$$

So max when $\theta = \frac{\pi}{4}$

$$A = \frac{(a^2 - b^2)^2 \sin\left(2 \cdot \frac{\pi}{4}\right)}{4ab}$$

$$= \frac{(a^2 - b^2)^2}{4ab}$$

$$P = \left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}} \right)$$

3	Correct Solution
2	Omitting 1 answer
1	Finding $\frac{dA}{d\theta} = \frac{(a^2 - b^2)^2 \cos 2\theta}{2ab}$

- (a) Given that α , β and γ are the roots to the equation $x^3 - x^2 + 5x - 3 = 0$, find the equation whose roots are $\alpha\beta$, $\alpha\gamma$ and $\beta\gamma$

Solution

$$P(x) = x^3 - x^2 + 5x - 3$$

$$\alpha\beta, \beta\gamma, \alpha\gamma = \frac{\alpha\beta\gamma}{\gamma}, \frac{\alpha\beta\gamma}{\alpha}, \frac{\alpha\beta\gamma}{\beta}$$

$$= \frac{3}{\gamma}, \frac{3}{\alpha}, \frac{3}{\beta} \quad \text{since } \alpha\beta\gamma = 3$$

$$\text{Let } x = \frac{3}{\alpha} \Rightarrow \alpha = \frac{3}{x}$$

$$P(\alpha) = \left(\frac{3}{x}\right)^3 - \left(\frac{3}{x}\right)^2 + 5\left(\frac{3}{x}\right) - 3 = 0$$

$$\frac{27}{x^3} - \frac{9}{x^2} + \frac{15}{x} - 3 = 0$$

$$x^3 - 5x^2 + 3x - 9 = 0$$

3	Correct Solution
2	Substituting $\alpha = \frac{3}{x}$
1	Finding $\alpha\beta\gamma = 3$

- (b) Let α be the complex root of the polynomial $z^7 = 1$ with the smallest possible argument.

$$\text{Let } \theta = \alpha + \alpha^2 + \alpha^4 \quad \text{and} \quad \phi = \alpha^3 + \alpha^5 + \alpha^6$$

- (i) Explain why $\alpha^7 = 1$ and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$

Solution

$$\text{Let } P(x) = z^7 - 1 = 0 \quad \text{since } \alpha \text{ is a root} \quad P(\alpha) = \alpha^7 - 1 = 0 \Rightarrow \alpha^7 = 1 \text{ by remainder theorem}$$

$$\text{Also } P(x) = z^7 - 1 = (z - 1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$$

Again since α is a root

$$P(\alpha) = (\alpha - 1)(\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1) = 0$$

$$\text{Since } \alpha \neq 1 \quad (\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1) = 0$$

2	Correct Solution
1	for one correct reason

(ii) Show $\theta + \phi = -1$ and $\theta\phi = 2$

Hence write a quadratic equation whose roots are θ and ϕ

Solution

$$\begin{aligned}\theta + \phi &= \alpha + \alpha^2 + \alpha^4 + \alpha^3 + \alpha^5 + \alpha^6 \\ &= \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 \\ &= -1 \quad \text{using pt (i)}\end{aligned}$$

$$\begin{aligned}\theta\phi &= (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) \\ &= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8 + \alpha^7 + \alpha^9 + \alpha^1 \\ &= \alpha^4 + \alpha^6 + 1 + \alpha^5 + 1 + \alpha\alpha^7 + 1 + \alpha^7\alpha^2 + \alpha^7\alpha^3 \\ &= \alpha^4 + \alpha^6 + 1 + \alpha^5 + 1 + \alpha + 1 + \alpha^2 + \alpha^3 \\ &= \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 + 3 \\ &= 2 \quad \text{using } \theta + \phi = -1\end{aligned}$$

A quadratic whose roots are θ, ϕ easiest quadratic would be monic

$$\theta + \phi = \frac{-b}{a} \quad \theta\phi = \frac{c}{a}$$

$$-1 = -b \quad 2 = c \quad \text{since } a = 1 \text{ with monic polynomial}$$

$$P(x) = x^2 + x + 2 = 0$$

3	Correct Solution
2	Finding 2 correct answers
1	Finding 1 correct answer

(iii) Show that $\theta = -\frac{1}{2} + \frac{i\sqrt{7}}{2}$ and $\phi = -\frac{1}{2} - \frac{i\sqrt{7}}{2}$

Solution

Solving $x^2 + x + 2 = 0$

$$\begin{aligned} x &= \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1} \\ &= \frac{-1 \pm \sqrt{-7}}{2} \\ &= \frac{-1 \pm \sqrt{7}i}{2} \end{aligned}$$

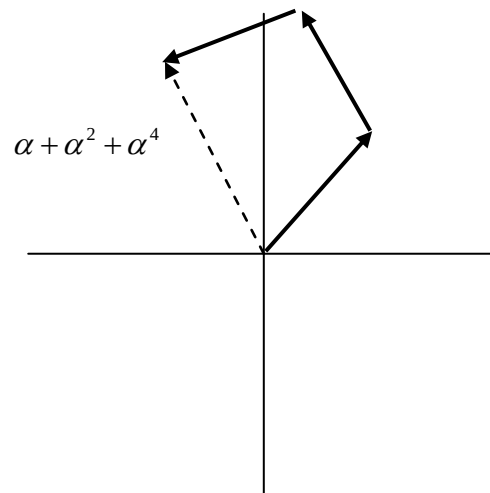
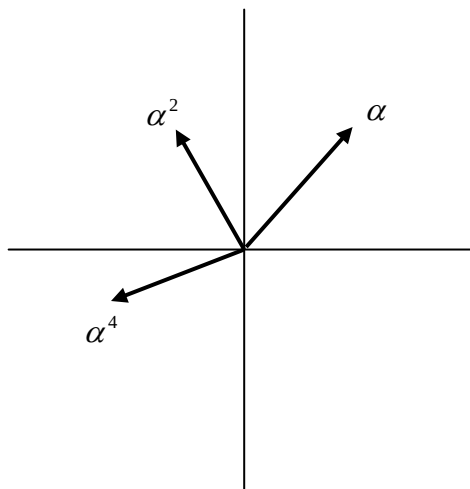
Determining whether $\theta = -\frac{1}{2} + \frac{i\sqrt{7}}{2}$ OR $\phi = -\frac{1}{2} + \frac{i\sqrt{7}}{2}$

Method 1 We can examine

$$\begin{aligned} \text{Im}(\alpha + \alpha^2 + \alpha^4) &= \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} \\ &> 0 \text{ from calculator} \end{aligned}$$

[found by solving $z^7 = 1$ using $\text{cis} \frac{2k\pi}{n}$]

Method 2 we can consider addition of vectors $\alpha + \alpha^2 + \alpha^4$ [found by solving $z^7 = 1$ using $\text{cis} \frac{2k\pi}{n}$]



Since $\alpha + \alpha^2 + \alpha^4$ has a positive argument

$$\theta = -\frac{1}{2} + \frac{i\sqrt{7}}{2} \text{ and } \phi = -\frac{1}{2} - \frac{i\sqrt{7}}{2}$$

2	Correct Solution
1	for finding $x = \frac{-1 \pm \sqrt{-7}}{2}$

(iv) Write α in modulus argument form and show

$$\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2} \quad \text{and} \quad \sin \frac{4\pi}{7} + \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$$

Solution

$$\alpha = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$$

$$\alpha^2 = \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7}$$

$$\alpha^4 = \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}$$

$$\theta = \alpha + \alpha^2 + \alpha^4$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} + \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}$$

$$= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \frac{\pi}{7} + i \sin \frac{2\pi}{7} + i \sin \frac{4\pi}{7} - i \sin \frac{\pi}{7}$$

Equating real & imaginary parts (using pt iii)

$$\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$$

$$\sin \frac{4\pi}{7} + \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$$

2	Correct Solution
1	for finding α

(c) The polynomial $P(z)$ is defined by $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$.

Solution

Given that $z - 2 + i$ is a factor of $P(z)$, express $P(z)$ as a product of real quadratic factors.

Factors are written in the form $(z - z_1)$

$$\text{So } z - 2 + i = [z - (2 - i)]$$

Since complex roots in a polynomial with real co-efficients occur in complex conjugates another factor is

$$= [z - (2 + i)]$$

Forming a quadratic factor

$$\begin{aligned} [z - (2 - i)][z - (2 + i)] &= [z - 2 + i][z - 2 - i] \\ &= z^2 - 4z + 5 \end{aligned}$$

Finding another factor

$$z^2 - 4z + 5 \overline{) z^4 - 2z^3 - z^2 + 2z + 10} \quad \begin{array}{r} z^2 + 2z + 2 \\ \hline \end{array}$$

So product of real quadratic factors

$$P(z) = (z^2 + 2z + 2)(z^2 - 4z + 5)$$

3	Correct Solution
2	for showing long division but with error
1	For finding $z^2 - 4z + 5$

(a) Consider the curve given by $5y - xy = x^2 - x - 2$

(i) Show that the curve has stationary points at $5 \pm 3\sqrt{2}$

Solution

$$5 \frac{dy}{dx} - x \frac{dy}{dx} - y = 2x - 1$$

$$(5 - x) \frac{dy}{dx} = 2x + y - 1$$

$$\frac{dy}{dx} = \frac{2x + y - 1}{5 - x}$$

Stationary pts at $\frac{dy}{dx} = 0$

$$\frac{2x + y - 1}{5 - x} = 0$$

$$2x + y - 1 = 0$$

$$y = 1 - 2x$$

Solving for x

$$5(1 - 2x) - x(1 - 2x) = (1 - 2x)^2 - (1 - 2x) - 2$$

$$5 - 11x + 2x^2 = x^2 - x - 2$$

$$x^2 - 10x + 7 = 0$$

$$x = \frac{10 \pm \sqrt{10^2 - 4 \cdot 1 \cdot 7}}{2 \cdot 1}$$

$$= 5 \pm 3\sqrt{2}$$

2	Correct Solution
1	for finding $\frac{dy}{dx} = \frac{2x + y - 1}{5 - x}$

- (ii) Explain why the curve approaches that of $y = -x - 4$ as $x \rightarrow \pm \infty$

Solution

$$5y - xy = x^2 - x - 2$$

$$(5 - x)y = x^2 - x - 2$$

$$y = \frac{x^2 - x - 2}{5 - x}$$

$$= -x - 4 + \frac{18}{-x + 5}$$

$$\begin{array}{r} -x-4 \\ -x+5 \overline{) x^2 - x - 2} \end{array}$$

$$\text{Now } \lim_{x \rightarrow \pm \infty} -x - 4 + \frac{18}{-x + 5} = -x - 4$$

2	Correct Solution
1	for working with partial error

- (b) For the hyperbola $\frac{x^2}{4} - \frac{y^2}{5} = 1$, find

- (i) The eccentricity.

Solution

$$e = \sqrt{\frac{b^2}{a^2} + 1}$$

$$= \sqrt{\frac{(\sqrt{5})^2}{(2)^2} + 1}$$

$$= \frac{3}{2}$$

1	Correct Solution
---	------------------

- (ii) The coordinates of the foci.

Solution

$$\text{Foci } (\pm ae, 0)$$

$$= \left(\pm 2 \cdot \frac{3}{2}, 0 \right)$$

$$= (\pm 3, 0)$$

1	Correct Solution
---	------------------

(iii) The equations of the directrices.

Solution

Directrices $x = \pm \frac{a}{e} \Rightarrow x = \pm \frac{4}{3}$

1 Correct Solution

(iv) The equations of the asymptotes.

Solution

$$y = \pm \frac{b}{a}x \Rightarrow y = \pm \frac{\sqrt{5}}{2}x$$

1 Correct Solution

(v) Sketch the hyperbola indicating the foci, the directrices and the asymptotes.

Solution

[Pending]

1 Correct Solution

(vi) Show that the point $P(2\sec\theta, \sqrt{5}\tan\theta)$ lies on the hyperbola and prove that the tangent to the hyperbola at P has the equation

$$\frac{x\sec\theta}{2} - \frac{y\tan\theta}{\sqrt{5}} = 1$$

Solution

Sub P into eqn. for hyperbola

$$\begin{aligned}\text{LHS} &= \frac{(2\sec\theta)^2}{4} - \frac{(\sqrt{5}\tan\theta)^2}{5} \\ &= \frac{4\sec^2\theta}{4} - \frac{5\tan^2\theta}{5} \\ &= \sec^2\theta - \tan^2\theta \\ &= \tan^2\theta + 1 - \tan^2\theta \\ &= 1 \\ &= \text{RHS}\end{aligned}$$

Showing the tangent is $\frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{5}} = 1$

$$\frac{x^2}{4} - \frac{y^2}{5} = 1$$

$$\frac{2x}{4} - \frac{2y}{5} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{5x}{4y}$$

At P $\frac{dy}{dx} = \frac{5 \sec \theta}{2\sqrt{5} \tan \theta}$

Equation of a line

$$y - \sqrt{5} \tan \theta = \frac{5 \sec \theta}{2\sqrt{5} \tan \theta} (x - 2 \sec \theta)$$

$$2\sqrt{5} \tan \theta y - 10 \tan^2 \theta = x 5 \sec \theta - 10 \sec^2 \theta$$

$$2\sqrt{5} \tan \theta y - 10 \tan^2 \theta = x 5 \sec \theta - 10 \tan^2 \theta - 10$$

$$5 \sec \theta x - 2\sqrt{5} \tan \theta y = 10$$

$$\frac{x \sec \theta}{2} - \frac{y \sqrt{5} \tan \theta}{5} = 1$$

$$\frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{5}} = 1 \quad \text{as } \frac{\sqrt{5}}{5} = \frac{1}{\sqrt{5}}$$

2	Correct Solution for showing pt lies on the hyperbola
1	

- (vii) If the tangent at P cuts the asymptotes at L and M , prove that $LP = PM$ and the area of triangle OLM is independent of the position of P .

Solution

Proving $LP = PM$

Finding co-ordinates of L & M

Sub asymptote $y = \pm \frac{\sqrt{5}}{2} x$ into the eqn for the tangent

$$\frac{x \sec \theta}{2} - \frac{\left(\frac{\sqrt{5}}{2}x\right) \tan \theta}{\sqrt{5}} = 1$$

$$\frac{x \sec \theta}{2} - \frac{x \tan \theta}{2} = 1$$

$$(\sec \theta - \tan \theta)x = 2$$

$$x = \frac{2}{(\sec \theta - \tan \theta)}$$

$$\text{and } y = \frac{\sqrt{5}}{(\sec \theta - \tan \theta)}$$

$$L\left(\frac{2}{(\sec \theta - \tan \theta)}, \frac{\sqrt{5}}{(\sec \theta - \tan \theta)}\right)$$

$$\frac{x \sec \theta}{2} - \frac{\left(-\frac{\sqrt{5}}{2}x\right) \tan \theta}{\sqrt{5}} = 1$$

$$\frac{x \sec \theta}{2} + \frac{x \tan \theta}{2} = 1$$

$$(\sec \theta + \tan \theta)x = 2$$

$$x = \frac{2}{(\sec \theta + \tan \theta)}$$

$$y = \frac{-\sqrt{5}}{(\sec \theta + \tan \theta)}$$

$$M\left(\frac{2}{(\sec \theta + \tan \theta)}, \frac{-\sqrt{5}}{(\sec \theta + \tan \theta)}\right)$$

Note finding the distance LP & LM was too troublesome (part marks may be awarded for those who tried)
Better to show P is the midpoint of LM

$$x = \frac{\frac{2}{(\sec \theta - \tan \theta)} + \frac{2}{(\sec \theta + \tan \theta)}}{2}$$

$$= \frac{1}{(\sec \theta - \tan \theta)} + \frac{1}{(\sec \theta + \tan \theta)}$$

$$= \frac{2 \sec \theta}{\sec^2 \theta - \tan^2 \theta}$$

$$= \frac{2 \sec \theta}{\tan^2 \theta + 1 - \tan^2 \theta}$$

$$= 2 \sec \theta$$

$$y = \frac{\frac{\sqrt{5}}{(\sec \theta - \tan \theta)} + \frac{-\sqrt{5}}{(\sec \theta + \tan \theta)}}{2}$$

$$= \frac{\sqrt{5}}{2} \left(\frac{1}{(\sec \theta - \tan \theta)} - \frac{1}{(\sec \theta + \tan \theta)} \right)$$

$$= \frac{\sqrt{5}}{2} \left(\frac{2 \tan \theta}{\sec^2 \theta - \tan^2 \theta} \right)$$

$$= \frac{\sqrt{5} \tan \theta}{\tan^2 \theta + 1 - \tan^2 \theta}$$

$$= \sqrt{5} \tan \theta$$

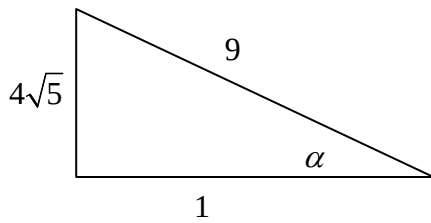
Proving OLM is independent of P

$$\tan \phi_1 = \frac{\sqrt{5}}{2} \quad \tan \phi_2 = \frac{-\sqrt{5}}{2}$$

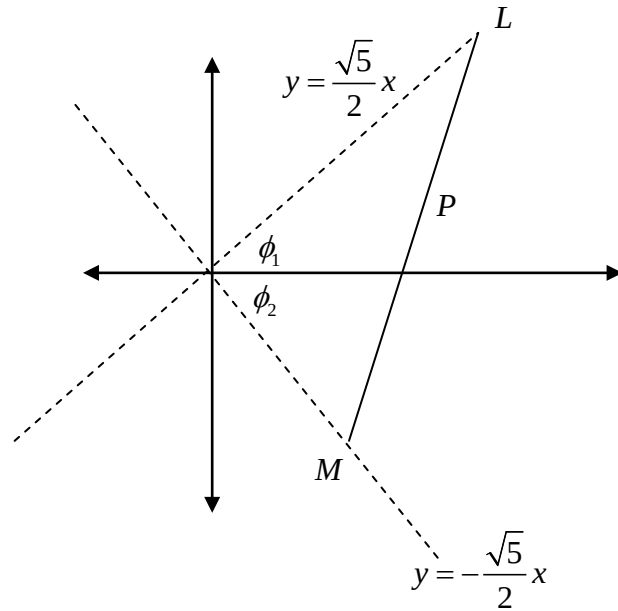
$$\phi = \phi_1 - \phi_2$$

$$\tan(\phi_1 - \phi_2) = \frac{\frac{\sqrt{5}}{2} - \frac{-\sqrt{5}}{2}}{1 + \frac{\sqrt{5}}{2} \cdot \left(-\frac{\sqrt{5}}{2}\right)}$$

$$= -4\sqrt{5} \Rightarrow \alpha > \frac{\pi}{2}$$



$$\sin \alpha = \frac{4\sqrt{5}}{9}$$



$$OL = \sqrt{\left(\frac{2}{\sec \theta - \tan \theta} - 0\right)^2 + \left(\frac{\sqrt{5}}{\sec \theta - \tan \theta} - 0\right)^2}$$

$$= \sqrt{\frac{9}{(\sec \theta - \tan \theta)^2}}$$

$$= \frac{3}{\sec \theta - \tan \theta}$$

$$OM = \sqrt{\left(\frac{2}{\sec \theta - \tan \theta} - 0\right)^2 + \left(\frac{-\sqrt{5}}{\sec \theta - \tan \theta} - 0\right)^2}$$

$$= \sqrt{\frac{9}{(\sec \theta - \tan \theta)^2}}$$

$$= \frac{3}{\sec \theta - \tan \theta}$$

$$A = \frac{1}{2} \cdot OL \cdot OM \cdot \sin \alpha$$

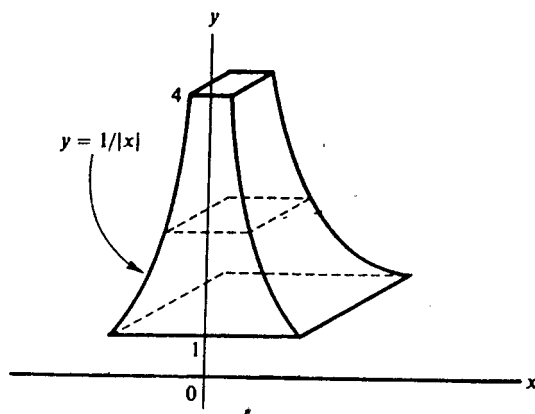
$$= \frac{1}{2} \cdot \frac{3}{\sec \theta - \tan \theta} \cdot \frac{3}{\sec \theta - \tan \theta} \cdot \sin \frac{4\sqrt{5}}{9}$$

$$= 2\sqrt{5}$$

- | | |
|---|--------------------------------------|
| 4 | Correct Solution |
| 3 | Finding $\sin \alpha$ or OL & OM |
| 2 | Finding the midpoint |
| 1 | Finding the co-ordinates of L & M |

Which is a constant term and therefore independent of P

- (a) The plan of a steeple is bounded by the curve $y = \frac{1}{|x|}$ and the lines $y = 4$ and $y = 1$.



Each horizontal cross-section is a square.

Find the volume of the steeple.

Solution

$$\delta V = (2x)(2x)\delta y$$

$$= 4x^2\delta y$$

$$= 4 \cdot \frac{1}{y^2} \cdot \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_1^4 \frac{4}{y^2} \delta y$$

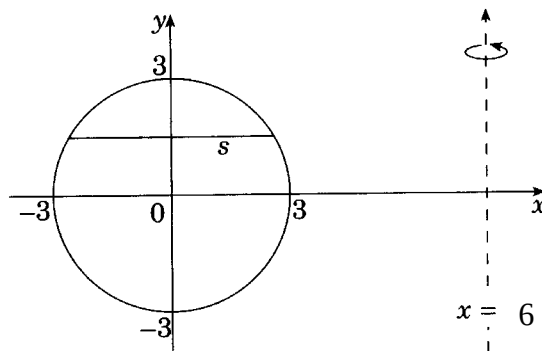
$$= \int_1^4 4y^{-2} dy$$

$$= \left[\frac{-4}{y} \right]_1^4$$

$$= 3 \text{ units}^3$$

4	Correct Solution
3	One arithmetic error with correct procedure
2	Finding $\delta V = 4 \cdot \frac{1}{y^2} \cdot \delta y$
1	For finding area of a slice

- (b) The circle $x^2 + y^2 = 9$ is rotated about the line $x = 6$ to form a ring.



- (i) When the circle is rotated, the line segment S at height y sweeps out an annulus.

Find the area of the Annulus.

Solution

$$\begin{aligned}
 A &= \pi(R^2 - r^2) \\
 &= \pi([6+x]^2 - [6-x]^2) \\
 &= \pi([6+x] - [6-x])([6+x] + [6-x]) \\
 &= 24\pi x
 \end{aligned}$$

2	Correct Solution
1	for showing pt lies on the hyperbola

- (ii) Hence find the volume of the ring

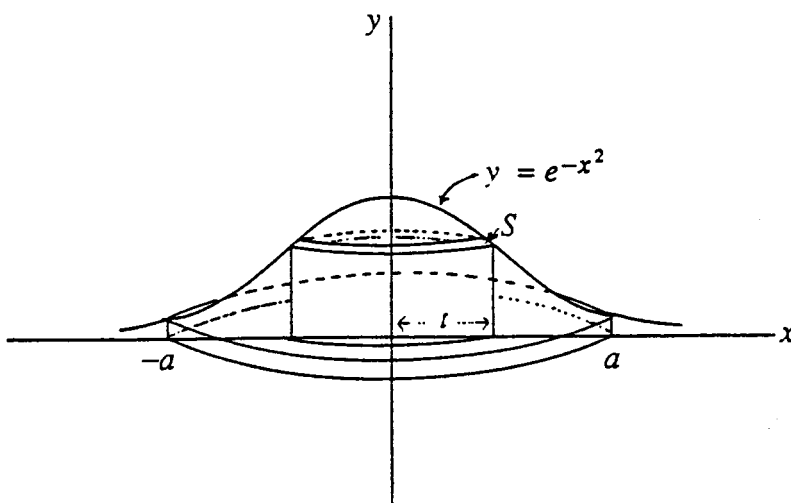
Solution

$$\begin{aligned}
 \delta V &= 24\pi x \delta y \\
 V &= \lim_{\delta y \rightarrow 0} \sum_{-3}^3 24\pi (\sqrt{9-y^2}) \delta y \\
 &= 24\pi \int_{-3}^3 \sqrt{9-y^2} dy \\
 &= 108\pi^2 \text{ units}^3
 \end{aligned}$$

3	Correct Solution
2	Showing $V = 24\pi \int_{-3}^3 \sqrt{9-y^2} dy$
1	Finding $\delta V = 24\pi x \delta y$

Note $\int_{-3}^3 \sqrt{9-y^2} dy$ is equal to the area of a semi-circle radius 3

- (c) The region under the curve $y = e^{-x^2}$ and above the x-axis is rotated about the y axis for $-a \leq x \leq a$ to form a solid as shown below.



- (i) Divide the resulting solid into cylindrical shells S of radius t as shown in the diagram and show each shell S has an approximate volume given by $\delta V = 2\pi t e^{-t^2} \delta t$, where δt is the thickness of the shell.

Solution

$$\begin{aligned}\delta V &= 2\pi r h \delta t \\ &= 2\pi(t)(e^{-t^2}) \delta t \\ &= \delta V = 2\pi t e^{-t^2} \delta t\end{aligned}$$

2	Correct Solution
1	For partial answer

- (ii) Hence calculate the volume of the solid.

Solution

$$\begin{aligned}V &= \lim_{\delta t \rightarrow 0} \sum_0^a 2\pi t e^{-t^2} \delta t \\ &= \pi \int_0^a 2t e^{-t^2} dt \\ &= -\pi \int_0^a -2t e^{-t^2} dt \\ &= -\pi \left[e^{-t^2} \right]_0^a \\ &= \pi (1 - e^{-a^2})\end{aligned}$$

2	Correct Solution
1	For some attempt to integrate

(iii) What is the limiting value of the volume of the solid as $a \rightarrow \infty$?

Solution

$$\text{As } a \rightarrow \infty, \quad e^{-a^2} \rightarrow 0$$

$$\text{So } \lim_{a \rightarrow \infty} \pi(1 - e^{-a^2}) = \pi$$

2	Correct Solution
1	For some attempt to integrate

Question 7.

(15 marks) Use a SEPARATE writing booklet.

(a) Let $I_n = \int_0^1 (1-x^2)^n dx$.

(i) Show by using integration by parts $I_n = \frac{2n}{2n+1} I_{n-1}$ for $n = 0, 1, 2, 3, \dots$

Solution

$$\begin{aligned} I_n &= \int_0^1 (1-x^2)^n dx &= \int_0^1 1 \cdot (1-x^2)^n dx \\ &= \left[x(1-x^2)^n \right]_0^1 - \int_0^1 x(n)(1-x^2)^{n-1}(-2x) dx \\ &= 0 - 2n \int_0^1 x^2(1-x^2)^{n-1} dx \\ &= -2n \int_0^1 (1-x^2-1)(1-x^2)^{n-1} dx \\ &= -2n \int_0^1 (1-x^2)(1-x^2)^{n-1} - 1(1-x^2)^{n-1} dx \\ &= -2n \int_0^1 (1-x^2)^n + 2n \int_0^1 (1-x^2)^{n-1} dx \end{aligned}$$

$$I_n = -2nI_n + 2nI_{n-1}$$

$$2nI_n + I_n = 2nI_{n-1}$$

$$I_n = \frac{2n}{2n+1} I_{n-1}$$

3	Correct Solution
2	For finding $-2n \int_0^1 (1-x^2)^n + 2n \int_0^1 (1-x^2)^{n-1} dx$
1	For finding $0 - 2n \int_0^1 x^2(1-x^2)^{n-1} dx$

(ii) Hence evaluate $\int_0^1 (1 - x^2)^4 dx$

Solution

$$I_n = \frac{2n}{2n+1} I_{n-1}$$

$$I_4 = \frac{8}{9} I_3$$

$$= \frac{8}{9} \left[\frac{6}{7} I_2 \right]$$

$$= \frac{8}{9} \left[\frac{6}{7} \right] \left[\frac{4}{5} I_1 \right]$$

$$= \frac{8}{9} \left[\frac{6}{7} \right] \left[\frac{4}{5} \right] \left[\frac{2}{3} I_0 \right]$$

$$= \frac{128}{315} \quad \text{as } I_0 = \int_0^1 1 dx = 1$$

3	Correct Solution
2	For using recurrence without evaluating last integral
1	For some evidence of correct procedure

(b) A special dish is designed by rotating the region bounded by the curve $y = 2 \cos x$ ($0 \leq x \leq 2\pi$) and the line $y = 2$ through 360° about the y axis.

i) Use the method of cylindrical shells to show that the volume of the dish is given by

$$4\pi \int_0^{2\pi} x(1 - \cos x) dx.$$

Solution

Each cylindrical shell has height $h=2-y$ and radius x .

$$\delta V = 2\pi r h \delta x$$

$$\delta V = 2\pi x (2 - y) \delta x$$

$$\delta V = 2\pi x (2 - 2 \cos x) \delta x$$

$$\text{Volume of dish} = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{2\pi} 2\pi x (2 - 2 \cos x) \delta x$$

$$= 4\pi \int_0^{2\pi} x (1 - \cos x) dx$$

3	Correct Solution
2	For showing $\lim_{\delta x \rightarrow 0} \sum_{x=0}^{2\pi} 2\pi x (2 - 2 \cos x) \delta x$
1	For showing $\delta V = 2\pi x (2 - y) \delta x$

ii) Hence find the volume.

Solution

$$V = 4\pi \int_0^{2\pi} x(1 - \cos x) dx$$

$$V = 4\pi \int_0^{2\pi} x dx - 4\pi \int_0^{2\pi} x \cos x dx$$

$$V = 4\pi \left[\frac{x^2}{2} \right]_0^{2\pi} - 4\pi \left\{ \left[x \sin x \right]_0^{2\pi} - \int_0^{2\pi} \sin x dx \right\}$$

$$V = 4\pi \left[\frac{x^2}{2} - x \sin x - \cos x \right]_0^{2\pi}$$

$$V = 4\pi \left[\frac{4\pi^2}{2} - 0 - 1 - (0 - 0 - 1) \right]$$

$$V = 8\pi^3 \text{ units}^3$$

3 Correct Solution

2 For showing $4\pi \left[\frac{x^2}{2} - x \sin x - \cos x \right]_0^{2\pi}$

1 For showing $4\pi \left[\frac{x^2}{2} \right]_0^{2\pi} - 4\pi \left\{ \left[x \sin x \right]_0^{2\pi} - \int_0^{2\pi} \sin x dx \right\}$

(c) The polynomial $P(x)$ is given by $P(x) = 2x^3 - 9x^2 + 12x - k$, where k is real.

Find the range of values for k for which $P(x) = 0$ has 3 real roots.

Solution

$P(x) = 2x^3 - 9x^2 + 12x - k$ will have 3 real roots when $P(x)$ has turning points on either side of the x axis.

Finding turning points:

$$P'(x) = 6x^2 - 18x + 12$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x+1) = 0$$

$$x = 2, -1$$

$$y = 4 - k, 5 - k$$

If turning points are on opposite sides $y_1 y_2 < 0$

$$(4 - k)(5 - k) < 0$$

$$4 < k < 5$$

3 Correct Solution

2 For finding the turning points and $y_1 y_2 < 0$

1 For finding the turning points

(a) Use integration by parts to find $\int \sin^{-1} x \, dx$.

Solution

$$\begin{aligned} \int \sin^{-1} x \, dx &= x \sin^{-1} x - \int x \cdot \frac{1}{\sqrt{1-x^2}} \\ &= x \sin^{-1} x - \int x(1-x^2)^{-\frac{1}{2}} \\ &= x \sin^{-1} x + \sqrt{1-x^2} + C \end{aligned}$$

3	Correct Solution
2	For finding $x \sin^{-1} x - \int x \cdot \frac{1}{\sqrt{1-x^2}}$
1	For finding one part of the IBP e.g. $x \sin^{-1} x$

$$\text{as } x(1-x^2)^{-\frac{1}{2}} = \int -f'(x) f(x)$$

(b) (i) Use De Moivre's Theorem to show that $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$

Solution

Let $z = \cos \theta + i \sin \theta$

$$\begin{aligned} z^4 &= (\cos \theta + i \sin \theta)^4 \\ &= \cos^4 \theta + 4\cos^3 \theta i \sin \theta + 6\cos^2 \theta i^2 \sin^2 \theta + 4\cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta \\ &= \cos^4 \theta + 4\cos^3 \theta i \sin \theta - 6\cos^2 \theta \sin^2 \theta - 4\cos \theta i \sin^3 \theta + \sin^4 \theta \\ &= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta + 4\cos^3 \theta i \sin \theta - 4\cos \theta i \sin^3 \theta \end{aligned}$$

Also by De Moivre's theorem

$$\begin{aligned} z^4 &= (\cos \theta + i \sin \theta)^4 \\ &= \cos 4\theta + i \sin 4\theta \end{aligned}$$

3	Correct Solution
2	For finding $\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$
1	For finding $z^4 = \cos 4\theta + i \sin 4\theta$

Equating real parts

$$\begin{aligned} \cos 4\theta &= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta \\ &= \cos^4 \theta - 6\cos^2 \theta (1 - \cos^2 \theta) + (\sin^2 \theta)(\sin^2 \theta) \\ &= \cos^4 \theta - 6\cos^2 \theta + 6\cos^4 \theta + (1 - \cos^2 \theta)(1 - \cos^2 \theta) \\ &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \end{aligned}$$

(iv) Show that the equation $16x^4 - 16x^2 + 1 = 0$ has roots

$$x_1 = \cos \frac{\pi}{12}, x_2 = -\cos \frac{\pi}{12}, x_3 = \cos \frac{5\pi}{12}, x_4 = -\cos \frac{5\pi}{12}$$

Solution

$$16x^4 - 16x^2 + 1 = 0$$

Let $x = \cos \theta$

$$16\cos^4 \theta - 16\cos^2 \theta + 1 = 0$$

$$2(8\cos^4 \theta - 16\cos^2 \theta + 1) - 1 = 0$$

$$2\cos 4\theta - 1 = 0$$

$$\cos 4\theta = \frac{1}{2}$$

$$4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \dots$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \dots$$

Now $x = \cos \theta$

$$x = \cos \frac{\pi}{12}, \quad x = \cos \frac{5\pi}{12}, \quad x = \cos \frac{7\pi}{12} = -\cos \frac{5\pi}{12}, \quad x = \cos \frac{11\pi}{12} = -\cos \frac{\pi}{12}$$

(iii) Hence show that $\cos \frac{\pi}{12} = \frac{\sqrt{2+\sqrt{3}}}{2}$

Solution

Solving $16x^4 - 16x^2 + 1 = 0$ by using quadratics

Let $m = x^2$ $16m^2 - 16m + 1 = 0$

$$m = \frac{16 \pm \sqrt{(-16)^2 - 4 \cdot 16 \cdot 1}}{2 \cdot 16}$$

3	Correct Solution
2	For finding $\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \dots$
1	For finding $2(8\cos^4 \theta - 16\cos^2 \theta + 1) - 1 = 0$

$$m = \frac{16 \pm 8\sqrt{3}}{32}$$

$$m = \frac{2 \pm \sqrt{3}}{4}$$

$$x^2 = \frac{2 \pm \sqrt{3}}{4}$$

$$x = \pm \sqrt{\frac{2 \pm \sqrt{3}}{4}}$$

Examining the positive roots as $\cos \frac{\pi}{12} > 0$

$$= \frac{\sqrt{2+\sqrt{3}}}{2} \quad \text{or} \quad \frac{\sqrt{2-\sqrt{3}}}{2}$$

Since $\frac{\sqrt{2+\sqrt{3}}}{2} > \frac{\sqrt{2-\sqrt{3}}}{2}$ and $\cos \frac{\pi}{12} > \cos \frac{5\pi}{12}$

$$\cos \frac{\pi}{12} = \frac{\sqrt{2+\sqrt{3}}}{2}$$

(c) $P(x)$ is a polynomial of degree n with rational coefficients.

If the leading coefficient is a_0 and $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are the roots of $P(x) = 0$ prove that:

$$P'(x) = \frac{P(x)}{x-\alpha_1} + \frac{P(x)}{x-\alpha_2} + \frac{P(x)}{x-\alpha_3} + \dots + \frac{P(x)}{x-\alpha_n}$$

Solution

$$P(x) = a_0(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)\dots(x-\alpha_n)$$

$$\log_e P(x) = \log_e a_0(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)\dots(x-\alpha_n)$$

$$\log_e P(x) = \log_e a_0 + \log_e(x-\alpha_1) + \log_e(x-\alpha_2) + \log_e(x-\alpha_3) + \dots + \log_e(x-\alpha_n)$$

$$\frac{P'(x)}{P(x)} = 0 + \frac{1}{(x-\alpha_1)} + \frac{1}{(x-\alpha_2)} + \frac{1}{(x-\alpha_3)} + \dots + \frac{1}{(x-\alpha_n)}$$

$$P'(x) = \frac{P(x)}{x-\alpha_1} + \frac{P(x)}{x-\alpha_2} + \frac{P(x)}{x-\alpha_3} + \dots + \frac{P(x)}{x-\alpha_n}$$

4 Correct Solution

3 For finding

2 For finding ?

1 For finding

$$P(x) = a_0(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)\dots(x-\alpha_n)$$